Development of a Coding System and Spatial Coordinate Framework for Bridge Structures Maintenance

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Abstract. With the advancement of technologies such as drones and image recognition, the automation level of bridge inspection and monitoring has gradually improved. However, the absence of an effective maintenance coding system for bridges hinders the component-level management of inspection results. Additionally, without a spatial coordinate system in traditional bridge maintenance, inspectors must manually match inspection data to the correct structural components, which slows down the process. To address these issues, this paper proposes a coding system and spatial coordinate framework for bridge structures maintenance. An algorithm for rapidly associating spatial coordinates with bridge components is further proposed, which enables automatic localization of detected defects during automated bridge inspections and significantly improves inspection efficiency.

Keywords: coding system, spatial coordinate, bridge, maintenance, algorithm

1 Introduction

In recent years, China has made remarkable progress in highway transportation infrastructure. According to the 2024 Statistical Bulletin on the Development of the Transport Sector, China now has 1.1081 million highway bridges. While the large-scale construction of bridges has supported rapid economic growth, it has also posed significant challenges for bridge maintenance. As bridges enter their mid-to-late service stages, structural defects have become increasingly severe, posing significant risks to the safe operation of bridges. In recent years, China has experienced frequent natural disasters, leading to repeated bridge collapses that seriously threaten public safety and property.

C. Ko et al. (eds.), *Proceedings of the 2025 8th International Conference on Civil Architecture, Hydropower and Engineering Management (CAHEM 2025)*, Advances in Engineering Research 293,
https://doi.org/10.2991/978-94-6239-600-5_28

Highway maintenance agencies typically monitor the safety condition of bridges dynamically through routine inspections and periodic assessments. Because bridge structures have a certain degree of spatial complexity, traditional inspection methods rely heavily on manual labor, resulting in low efficiency. With the emergence of a new wave of technological revolution and industrial transformation, rapid advances in technologies such as the Internet of Things (IoT), drones, and image recognition have significantly enhanced the convenience of bridge inspection and monitoring.

Currently, research on intelligent bridge inspection primarily focuses on intelligent inspection equipment and image-based defect recognition [1-3]. M. Hao proposed a method for automatic detection and 3D reconstruction of bridge cracks [4]. The automatic detection model of bridge cracks was optimized to further improve the robustness. H. Wan proposed a computer vision-based method for vehicle-induced bridge deflection measurement and vehicle localization [5]. The proposed computer vision-based deflection measurement and vehicle localization method combines non-contact measurement and high precision, offering reliable data support for rapid assessment of small and medium-span bridges. L. OU introduces an automated UVA inspection system featuring with deployed UAV docking station, remote automated inspection, AI image recognition and structural diseases digital twins [6]. The automated drone inspection system has been officially deployed. It achieves 100% coverage of critical bridge components, including steel structures, arches, girders, hangers, bearings, and piers. Automatic linkage has not yet been implemented. X. Xiong converted drone-captured images into a 3D reality model [7]. Crack features such as area, length, and width were extracted and processed. The specific location and dimensions of the cracks can then be intuitively displayed in the 3D model. The 3D model and the spatial location of cracks are displayed together. This enables realistic visualization of the cracks. L. Zhao demonstrated that UAV oblique photogrammetry can yield relatively accurate inspection results [8]. The above research indicates that automated daily inspections and periodic checks of bridges are gradually being adopted.

Current research on UAV inspection images focuses on defect identification and crack quantification. However, achieving automated and accurate localization of local defect images in bridge 3D maintenance models remains a significant challenge. At present, the automatic localization of images in the 3D point cloud model of the bridge is mainly achieved through oblique photogrammetry 3D reconstruction [9-10]. C. Fu proposed a coordinate projection and mapping method to map the defect image coordinates from GPS WGS-84 coordinate system to BIM model coordinate system [11]. the proposed method can realize real-time automated accurate localization of bridge defects on BIM model. Y. Li proposed an automated bridge damage detection and localization method based on deep learning and Structure from Motion (SfM) technology [12]. The method realized the complete process from image recognition to the accurate mapping of damage information in Building Information Model (BIM). However, due to the lack of an effective spatial coding system for bridge maintenance, the precise location of defects cannot be directly correlated with bridge components.

The precise classification of bridge structures and the standardization of bridge defect data are key to a digital bridge maintenance system. There has been considerable research on bridge coding systems, which are generally formulated in accordance with the relevant requirements for the assessment of the technical condition of highway bridges. K. Song proposed a refined structural coding system and a standardized method for defect data of long-length bridges [13]. This system divides long-length bridges into a five-level structural hierarchy: bridge, section, member, component, and sub-component. It provides structural support for establishing a refined and high-quality maintenance system for long bridges. Q. Meng summarized a classification structure for bridge physical elements, along with a corresponding hierarchical coding scheme and coding content [14]. A logical method was developed to transform bridge component information into semantic bridge codes based on "spatial relationships" and "dependency relationships". These coding systems facilitate structured management of bridge maintenance data. These coding systems contribute to the structured management of bridge maintenance data. However, in the absence of a spatial coordinate framework, automated association between spatial location data and structural components remains challenging.

In response to emerging technologies and evolving requirements in bridge inspection and monitoring, this paper proposes a spatial coding system for bridges. By integrating conventional bridge structural coding with spatial positioning, the system enables automatic association between spatially located defects and corresponding bridge components. The implementation of this coding framework facilitates automated, component-level data management.

2 Spatial Coordinate Coding System for Bridge Components

This section primarily presents the coding principles, coordinate system configuration, and geometric data storage specifications of the spatial coordinate coding system for bridge components.

2.1 Coding Principles

Taking full account of the application needs throughout the entire life cycle of highways, a study was conducted to formulate bridge component division and coordinate rules. This method should include at least the following key elements:

- Bridges should be disassembled according to the fineness of the components to meet the specific requirements of maintenance management and inspection evaluation;
- Component numbering should be based on the regular inspection requirements;
- The coordinate system should include the specific spatial positioning information of each component, expressing the location of the component in a spatial volume manner to support UAV applications.

2.2 Coordinate System and Coding Rules

During the R&D phase, this project adopts the WGS84 coordinate system due to its native compatibility with high-precision GNSS devices, ensuring positioning accuracy and facilitating multi-source data integration. Based on the above principles, the bridge spatial coordinate system shown in Table 1 is established. This coding system extends the conventional bridge coding framework by incorporating spatial positioning information. The spatial locations of various components are recorded using a spatial bounding cuboid method, as illustrated in Fig. 1.

Table 1. Type Styles

Type	Coordinate	Example	Description
Mileage-based Polar Coordinates	Bridge Code	G103110112L0070	Adopt the existing bridge coding system directly.
	Route ID	G103	Adopt the route number.
	Section Type	3	Indicate the section type, including uphill, downhill, bidirectional, or ramp.
	Start Station	15.440	The starting station of the subgrade or pavement segment.
	End Station	15.450	The ending station of the subgrade or pavement segment.
	Lateral Boundary 1 Location	4.2	Positioning parameter of the boundary closer to the road centerline.
	Lateral Boundary 2 Location	7.7	Positioning parameter of the boundary farther from the road centerline.
	Vertical Boundary 1 Location	0	Vertical distance from the pavement to the nearer boundary.
	Vertical Boundary 2 Location	-0.2	Vertical distance from the pavement to the farther boundary.
	Component ID	12-3#	**Component Positioning Principles**: Each component is uniquely identified by a combination of longitudinal and transverse codes. • **Longitudinal Coding**: Assigned sequentially in the direction of increasing station numbers. • **Transverse Coding**: For each separated structural unit, numbering proceeds from the outermost edge toward the road centerline, with the centerline serving as the reference axis. For superstructure components—excluding bearings, pier/abutment cap beams, and deck-level elements such as expansion joints—the unique code is formed as: “Span Number + Longitudinal Code + Transverse Code.” For substructure components, including bearings, pier/abutment cap beams, and expansion joints, the unique code is formed as:

Type	Coordinate	Example	Description
			"Pier/Abutment Number + Longitudinal Code + Transverse Code."
	Component Type	Box Girder	Select according to the specifications.
Absolute Coordinates	Centerline Start Coordinates	(L1, B1, H1)	The road centerline coordinates corresponding to the end of the component with the smaller station number.
	Centerline End Coordinates	(L2, B2, H2)	The road centerline coordinates corresponding to the end of the component with the larger station number.
	Centroid Coordinates	(L3, B3, H3)	The geometric center coordinates of the component.

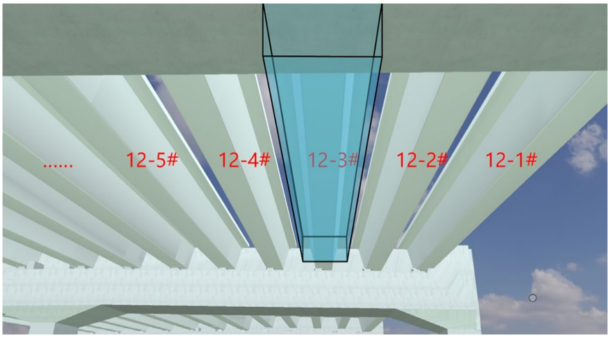

Fig. 1. Example of Rectangular Spatial Envelope for a T-Girder Component

The spatial reference for bridge component positioning is the road centerline. The precise coordinates of the centerline are recorded using three parameters: Centerline Start Coordinates, Centerline End Coordinates, and Centroid Coordinates. Each coordinate is expressed in three components—longitude (L), latitude (B), and elevation (H)—formatted as (L, B, H). The specific coordinate system used may be adapted according to local or regional standards. Centroid Coordinates represent the longitude, latitude, and elevation of the geometric center of the bridge component's spatial cell. This coordinate triplet accurately reflects the real-world location of the bridge component and can be used for rapid spatial positioning and retrieval.

2.3 Geometric Data Storage Specification

It is recommended to store the geometric data of bridge cells in GeoJSON format. Each storage object should include three main elements: **type**, **properties**, and **geometry**.

- The **type** field for all bridge component cells shall be set to "Feature".
- The **properties** field shall include all parameter information listed in Table 1, encompassing component identification, structural classification, coding details, and other relevant business attributes.

- The **geometry** field shall represent the planar projection of the bridge component using a two-dimensional polygon. The geometry type shall be specified as "Polygon".

3 Bridge Coordinate Transformation Method

Based on the bridge coordinate system proposed in this paper, it is now possible to automatically locate the specific structural component associated with an inspection-identified defect using its spatial longitude, latitude, and elevation coordinates.

After all bridges in the region have been decomposed into components and stored spatially, the number of components becomes extremely large, necessitating optimization of the computation workflow. An efficient positioning calculation method is illustrated in Fig. 2. The main computational steps include:

- Identify defects using drones and obtain their spatial coordinates through 3D reconstruction or image-based depth estimation.
- Filter out components whose centroid coordinates (longitude, latitude, and elevation) exceed a specified threshold. This operation is performed directly within the database to improve computational efficiency.
- For nearby components, obtain the spatial bounding envelope of the component by combining the absolute position of the lane centerline with the component's relative bounding envelope. Then, use spatial algorithms to determine whether the spatial location lies within the component's bounds.
- Retrieve information such as the bridge component's code, chainage, and component type, and output the matching results. Link the defect data to the component codes.

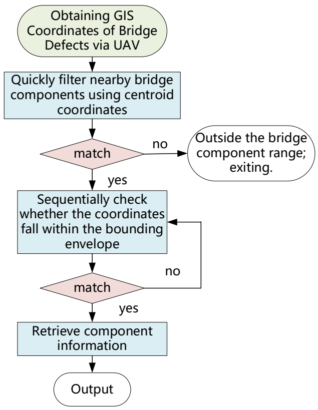

Fig. 2. Typical spatial coordinate positioning calculation workflow

The proposed algorithm efficiently links detected defects to the correct bridge components using spatial coordinates. By combining centroid-based filtering with precise envelope checks, it quickly identifies the right component and connects defect data to component codes for better maintenance and management.

4 Applicability and Limitations

The drone inspection system works well on small-span bridges (e.g., beam or slab bridges), where flight is stable and GNSS coverage is sufficient for accurate georeferencing. For large or complex bridges, challenges include GNSS signal loss under decks or near towers, limited maneuverability in tight spaces, and wind interference. These factors can reduce positioning accuracy and imaging quality, requiring customized flight paths or alternative localization methods.

In drone-based bridge inspection, GNSS signals are often degraded or lost under the bridge deck due to signal blockage from the structure itself, leading to significant positioning drift. This results in spatial misalignment of detected defects, with potential location errors ranging from several decimeters to over a meter in severe cases. Extensive field trials under varied bridge geometries and environmental conditions are needed to quantify and mitigate these errors.

5 Conclusions

This paper presents a spatial coordinate coding system for bridge components that integrates structural coding with precise spatial positioning. By introducing centroid coordinates and spatial bounding envelopes, the proposed framework enables automatic and efficient association between defect locations and their corresponding bridge components. The accompanying algorithm significantly reduces manual effort in inspection data processing, supports component-level maintenance management, and enhances the overall efficiency and accuracy of automated bridge inspections. This approach provides a practical foundation for digital, intelligent bridge management.

Acknowledgments

Funding provided by Beijing Transportation Industry Science and Technology Project (BJJTW-2024-ZHJT-17).

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